

SHeF-WIP: Walking-in-Place based on Step Height and Frequency for Wider Range of Virtual Speed

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ABSTRACT

Walking-in-place (WIP) approaches face difficulties in reaching high locomotion speeds because of the required high step frequency, rapidly creating an awkward or risky experience for the user. In this paper, we introduce a novel WIP approach called Step-Height-and-Frequency (SHeF) WIP, which considers a second parameter, i.e., the step height, in addition to the step frequency, to better control the speed of advancement. We compared SHeF-WIP with a conventional WIP system in a user study conducted with 12 participants. Our results suggest that SHeF-WIP enabled them to reach higher virtual speeds (+80%) with more efficacy and ease.

Index Terms: Computer Graphics—Three-Dimensional Graphics and Realism—Virtual Reality—;

1 INTRODUCTION

Walking-in-place (WIP) is a locomotion technique that enables users to "walk through" vast virtual environments (VEs) within a limited physical space. Conventional WIP techniques utilize the step frequency to achieve low latency and natural control of virtual walking speed based on a gait behavior analysis [3]. Although WIP has potential, one of the largest problems is the limited range of the stably achieved virtual speed. In general, the maximum controllable speed of WIP is around 2.5 m/s, and WIP faces difficulties in achieving a higher speed, that is, a "running" speed [1, 3]. To reach a faster speed, the more intensive stepping the users have to apply. However, intensive stepping increases not only the fatigue of the user but also the risk of injury and the shaking of the head-mounted display, which decreases the feeling of immersion in the VR environment and results in an unstable tracking of the sensors on the feet.

2 SHEF-WIP: A NOVEL WIP APPROACH

We propose a novel WIP method called step-height-and-frequency (SHeF) WIP that utilizes both the step height and step frequency. We assumed that the addition of a new parameter will provide additional freedom to the user to control the walking speed, leading to a higher speed with more ease and stability.

SHeF-WIP is based on the gait-understanding-driven (GUD) WIP [3] that is one of the most conventional WIP approach. The GUD-WIP utilizes the biomechanics of the human gait to understand the stepping gestures and calculates the virtual speed using the step frequency. Therefore, it can create a walking speed that matches the speed at the same frequency as actual walking. SHeF-WIP introduces a step height factor into the GUD-WIP approach. Equation (1) shows the GUD-WIP algorithm and Equation (2) shows our SHeF-WIP algorithm:

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$$|v_g| = \left(\frac{f}{1.57} \times \frac{H}{1.72} \right)^2 \quad [m/s] \quad (1)$$

$$|v_s| = |v_g| \times \frac{sh}{0.1} \quad [m/s] \quad (2)$$

where $|v_g|$ is the virtual speed of GUD-WIP, $|v_s|$ is the virtual speed of SHeF-WIP, f is the step frequency, H is the user's height, and sh is the step height.

The GUD-WIP algorithm refers to the relationship between the real walking speed, step frequency, and subject height as proposed by Dean [2]. GUD-WIP recognizes the user's stepping gestures, separating one-step gestures for three states (grounded, ascending, and descending), and therefore can estimate the step frequency with only a fraction of a step completed. Our SHeF-WIP approach adopts the same method of calculating the step frequency as GUD-WIP by recognizing the stepping gesture.

The SHeF-WIP algorithm adds a step height factor to the GUD-WIP algorithm (Equation (2)). Here, 0.1 m indicates the standard step height obtained by our pilot test. As we can see in Equation (2), the virtual speed follows the GUD-WIP algorithm while users maintain a 0.1 m step height, and is amplified/decreased depending on a higher/lower step height. This means that the speed of SHeF-WIP is two- or three-times the speed of GUD-WIP when the user's step height is 0.2 or 0.3 m.

3 USER STUDY

We conducted an experiment (10 males and 2 females, 11 of whom were in their 20s and 1 was in her 30s) to compare the range of walking speeds that can be achieved with GUD-WIP and SHeF-WIP. The hypotheses of the experiment was that SHeF-WIP can achieve a wider range of speed with more stability than GUD-WIP.

The main experimental task required participants to chase a ball for 20s in VR with the goal of matching its speed. The ball's speed was randomly chosen each trial from 0.5, 1.0, 1.5, 2.0, 3.5, 4.5 m/s. The collected data was the average stepping frequency, the average step height, the average distance between the ball and an indicator object of user's position/speed, and the walking speed. The participants were asked to answer questionnaires about controllability, sense of walking, realism of walking, tiredness, cybersickness, and preference of the system. After all trials, the participants were asked about their preference for the WIP techniques and their reasoning.

4 RESULTS AND DISCUSSION

Fig. 1 shows the result of the average distance from the center of the ball to that of the indicator of the participant's position/speed. Lower values of the average distance means that participants were able to follow the target more accurately. A two-factor analysis of variance (ANOVA) was conducted. A significant main effect was observed in both the WIP factor ($F(1, 11) = 15.90, p = 0.0021, \eta^2 = 0.048$) and speed factor ($F(1.59, 17.54) = 639.09, p < 0.001, \eta^2 = 0.372$). A significant interaction effect was also found ($F(1.91, 21.03) = 101.18, p = 0.014, \eta^2 = 0.071$). Then, a post-hoc test for the effect of the WIP factor at each speed revealed that the average distance of

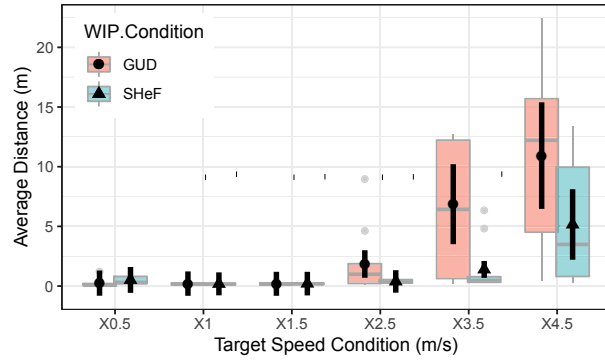


Figure 1: Average distance from the center of the ball to that of the indicator of the participant's position/speed

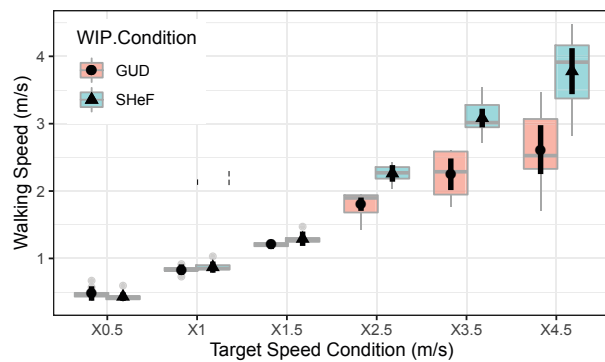


Figure 2: Average walking speed

SHeF-WIP was significantly lower than that of GUD-WIP under the speed conditions of 3.5 and 4.5 m/s.

Fig. 2 indicates the average walking speed during the chasing task. A two-way ANOVA was performed and a significant main effect was observed for both the WIP factor ($F(1, 11) = 80.56, p < 0.001, \eta^2 = 0.036$) and the speed factor ($F(1.6, 17.61) = 395.68, p < 0.001, \eta^2 = 0.872$). In addition, a significant interaction effect was found ($F(2.09, 22.95) = 36.92, p < 0.001, \eta^2 = 0.042$). The post-hoc tests for the interaction effect suggests that the walking speed of SHeF-WIP was significantly faster than that of GUD-WIP at the speed of 1.5, 2.5, 3.5, and 4.5 m/s. This result shows that participants were able to achieve a walking speed closer to the target speed with the SHeF-WIP when moving at moderate and high speeds.

Fig. 3 shows the standard deviation (sd) of individual walking speed during each chasing task, which indicates how stably the participants could keep a constant walking speed. As a result of a two-factor ANOVA, a significant main effect was observed in both WIP factor ($F(1, 11) = 7.81, p = 0.017, \eta^2 = 0.026$) and speed factor ($F(2.21, 24.27) = 60.07, p < 0.001, \eta^2 = 0.616$), and a significant interaction effect was found ($F(2.57, 28.29) = 5.68, p = 0.005, \eta^2 = 0.052$). The post-hoc test for the interaction effect suggests that the sd of the walking speed of SHeF-WIP is significantly smaller than that of GUD-WIP at the speed condition of 2.5 and 3.5 m/s. This result shows that participants were able to control their walking speed more precisely with the SHeF-WIP at high speeds.

These results support our hypothesis. As we can see in Fig. 1, the participants successfully achieved a speed of 3.5 m/s with SHeF-WIP, and participants started to face difficulties in catching up with the target at 2.5 m/s with GUD-WIP. The similar conclusion is

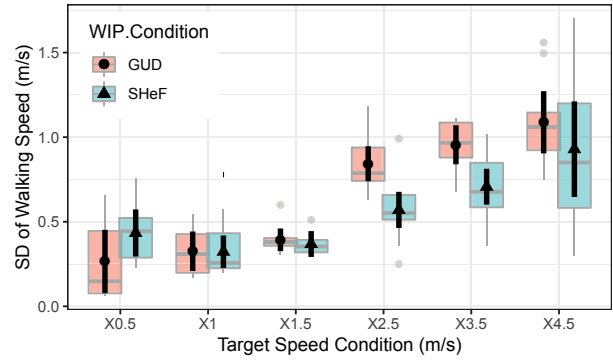


Figure 3: The standard deviations of the individual walking speed during each chasing task. Note that this standard deviation does not indicate that of Fig. 2. These results show how stably the participants maintain a constant speed.

obtained from the results of Fig. 3, because the speed control of WIP is significantly more stable with SHeF-WIP than with GUD-WIP at the speed condition of 2.5 and 3.5 m/s. Considering the results of Fig. 1 and Fig. 3, we can roughly estimate the ceiling of virtual speed that users can control with stability is the speed at 2.5 m/s condition for GUD-WIP and the speed at 3.5 m/s condition for SHeF-WIP. These speeds are 1.84 ± 0.18 m/s for GUD-WIP and 3.10 ± 0.23 m/s for SHeF-WIP based on Fig. 2. In addition, considering that SHeF-WIP also enabled stable control of lower speed as GUD-WIP (Fig. 2), we can conclude that SHeF-WIP can achieve wider range of virtual speed (about +80%) with more stability than GUD-WIP. As a subjective feedback of the participants, some participants felt more comfortable to adjust the speed (4 out of 12) or more natural and real (4 out of 12) with SHeF-WIP, while others argued the easiness of control with GUD-WIP (5 out of 12).

5 CONCLUSION

In this paper, we introduced a novel WIP technique called SHeF-WIP, which relies on both the step height and frequency for a wider range of virtual walking speeds. To evaluate SHeF-WIP, we compared it with GUD-WIP technique in a user study conducted with 12 participants. The participants were asked to walk in VE while maintaining a constant virtual speed under various speed conditions. The results suggest that SHeF-WIP enabled the participants to reach higher speed increases of up to +80% compared with GUD-WIP while achieving greater stability and ease. Therefore, we believe that SHeF-WIP is a promising WIP-based locomotion technique for VR applications requiring a wider range of advancing speeds.

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REFERENCES

- [1] L. Bruno, J. Pereira, and J. Jorge. A new approach to walking in place. In *IFIP Conference on Human-Computer Interaction*, pp. 370–387. Springer, 2013.
- [2] G. Dean. An analysis of the energy expenditure in level and grade walking. *Ergonomics*, 8(1):31–47, 1965.
- [3] J. D. Wendt, M. C. Whitton, and F. P. Brooks. Gud wip: Gait-understanding-driven walking-in-place. In *2010 IEEE Virtual Reality Conference (VR)*, pp. 51–58, 2010. doi: 10.1109/VR.2010.5444812